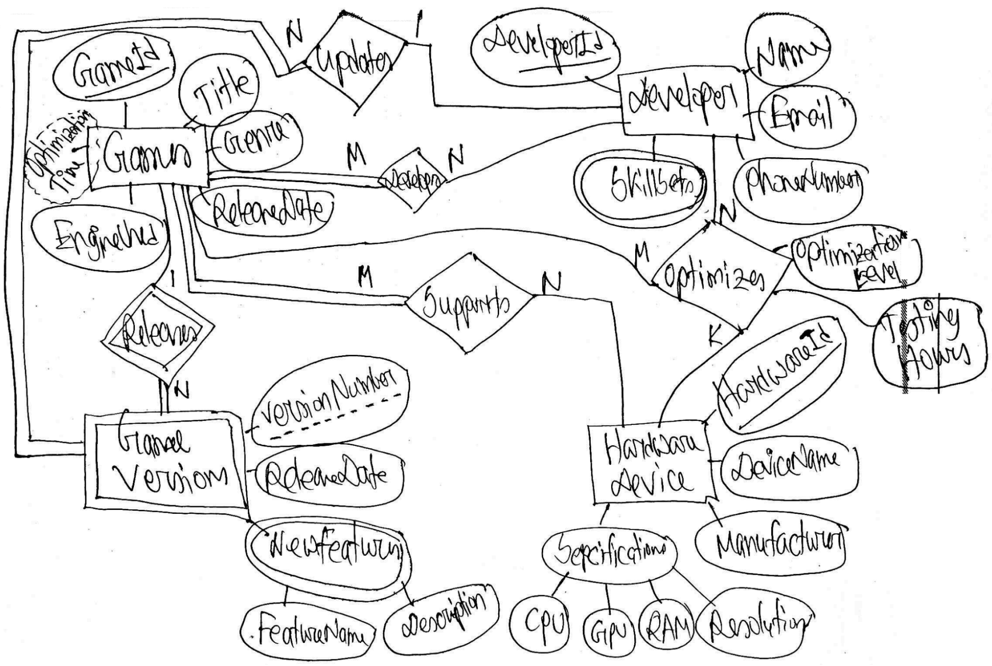
CSE 370– Database Systems

QUIZ 1

A tech company is developing a new database system called the "Virtual Reality Game Development System" to manage its game development projects. **Construct** an ER Diagram which fulfills the following data requirements.

- The company is developing virtual reality games. Each game is identified by a unique GameID. A game also has a Title, a Genre, ReleaseDate, and EngineUsed attributes.
- A game must be developed by at least one developer, but a game can have multiple developers developing it. Each developer is identified by a unique DeveloperID and has a Name, Email, PhoneNumber, and multiple SkillSets. A developer may develop several games.
- A Game must support different hardware devices, and a hardware device supports multiple games. Each hardware device has a unique HardwareID. A hardware device also has a DeviceName, Manufacturer, and Specifications which is composed of CPU, GPU, RAM, and Resolution.
- Games are released as different Game Versions. Game versions have the following attributes: VersionNumber, ReleaseDate and multiple NewFeatures which is composed of FeatureName and Description. Version numbers are the same for different games (e.g. 1, 2, 3 and so on). A developer can update multiple game versions; each game version is updated by one developer.
- A developer can optimize a game for specific hardware devices to ensure the best user experience. OptimizationLevel and the TestingHours spent on optimization are recorded for each optimization.
- Additionally, the total OptimizationTime for a game can be calculated by summing the testing hours contributed by all developers optimizing that game. This value will not be stored but should be shown using the appropriate ER symbol.

Do not assume any additional attributes, entities, relationships, key, multivalued, or composite attributes other than those mentioned above. For participation constraints and cardinality ratios, if they are not specified, you may assume them according to logical reasoning.



A tech startup is developing a new recipe sharing platform called "Chef's Corner" and needs a database system designed to manage it. **Construct** an ER Diagram which fulfills the following data requirements.

- Each Chef is identified by a unique ChefID. Chefs also have a Name, email address, multiple SocialMedia which are composed of MediumName and userHandle, ExperienceLevel and multiple CuisineSpecializations.
- Chefs create Recipes, each identified by a unique RecipeID. Each recipe has a Title, CuisineType, DishType, PreparationTime, and CookingTime. A recipe must be created by one chef, but a chef can create multiple recipes.
- The CookingTime of a recipe can be calculated by summing the durations of all its recipe steps described below. This value is not stored in the database but must be shown using the appropriate ER symbol.
- Each recipe consists of several Recipe Steps. Recipe Steps have the following attributes: StepNumbers, Description, and Duration. Step numbers are the same for each recipe (e.g. 1, 2, 3,...,and so on).
- Recipe steps may include several Ingredients, and an ingredient can be included in several recipe steps. Each Ingredient is identified by a unique IngredientID and also has IngredientName, Category, and NutritionalInformation which is composed of

Calories, Fat, and Protein. The Measurement and AdditionalInstruction for each ingredient put in each recipe step is recorded.

- Chefs teach selected Recipes in Cooking Classes. Each class has a unique ClassID and other attributes such as ClassName, Date, and Location.

Do not assume any additional attributes, entities, relationships, key, multivalued, or composite attributes other than those mentioned above. For participation constraints and cardinality ratios, if they are not specified, you may assume them according to logical reasoning.